

Readme Contents

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1. What is this?

The Bcalc Recoil Calculator is an open-source firearm recoil calculator.

You can enter specific firearm and ammo data and the calculator will estimate:

- free recoil energy: total energy absorbed
- recoil velocity: speed the firearm moves backwards
- average impulse force: average force on hands/ shoulder
- perceived recoil score: to be used in recoil comparisons

An internet connection is not required once installed. No user data is transmitted.

2. Who is this for?

Anyone interested in comparing recoil related calculations of different firearm and ammunition combinations.

3. Basic Linux Installation Guide

1. Install System Dependencies

- OpenMandriva: `sudo dnf install python3 python3-pip tkinter`
- Vendeoul Wolf: `sudo apt install python3 python3-pip python3-tk python3-venv`

2. Create and Activate Python Virtual Environment

- `mkdir -p ~/projects/bcalcrecoil`
- `cd ~/projects/bcalcrecoil`
- `python3 -m venv venvrecoil`
- `source venvrecoil/bin/activate`

3. Install Python Packages

- `pip install -r requirements.txt`

4. Download the Files

- download the applications files and place them in `~/projects/bcalcrecoil` (all .py, .csv, .png, .desktop files)

5. Run the Application

- from the activated venvrecoil environment: `python3 startrecoil.py`

Optional: Launch Icon

1. Copy to applications folder:

- `cp bcalcrecoil.desktop ~/.local/share/applications/`

2. Copy icon

- `mkdir -p ~/.local/share/icons`
- `cp bcalcrecoilicon.png ~/.local/share/icons/`

3. Refresh desktop menu: search "Bcalc Recoil Calculator" in your app menu (should see the quail & lightning bolt image)

4. User Guide

Step 1. Enter a unique name for the data set if you want to be able to refer to it later without having to enter all the variables again. Not required however.

The screenshot displays the Bcalc Recoil Calculator interface, which is divided into two main sections: Input Parameters and Results.

Input Parameters:

- Unique Name:** 308 WIN
- Cartridge:** 308 Win
- Firearm Class:** rifle: standard
- Action Type:** bolt action
- Gun Weight:** 8.0 lbs
- Bullet Weight:** 125 grains
- Muzzle Velocity:** 2973 fps
- Powder Charge:** 46.5 grains
- Muzzle Device:** none
- Stock Fit:**
 - LOP:** just right
 - Comb Height:** perfect cheek weld
 - Recoil Pad:** soft recoil pad

Results:

	Data Set 1	Data Set 2	Data Set 3
Unique Name:	308 WIN	--	--
Cartridge Selected:	308 Win	--	--
Firearm Class:	rifle: standard	--	--
Action Type:	bolt action	--	--
Gun Weight:	8.0 lbs	--	--
Bullet Weight:	125.0 gr	--	--
Muzzle Velocity:	2973 fps	--	--
Powder Charge:	46.5 gr	--	--
Muzzle Device:	none	--	--
LOP:	just right	--	--
Comb Height:	perfect cheek weld	--	--
Recoil Pad:	soft recoil pad	--	--

Calculated Results:

Free Recoil Energy	14.92 ft-lbs	-- ft-lbs	-- ft-lbs
Recoil Velocity	10.96 fps	-- fps	-- fps
Avg Impulse Force	1434 lbf	-- lbf	-- lbf
Perceived Recoil Score	5.7 /10	-- /10	-- /10

At the bottom of the input section, there are two buttons: **Calculate** (blue) and **Save Data Set** (green). In the results section, there is a small illustration of a chicken and a blue button with a hamburger menu icon.

Step 2. Select cartridge, firearm class, and action type from the drop-down menus.

Step 3. Enter gun weight (lbs), bullet weight (grains), and muzzle velocity.

Step 4. Enter a powder charge weight (grains) if known. The calculator will estimate if left blank.

Step 5. Select muzzle device, LOP, comb height, and recoil pad from the drop-down menus.

Step 6. Press the blue Calculate button to kick off the calculator. The results show up in the right panel.

Step 7. If you wish to compare a data set, change the input parameters, and press the Calculate button. In this example, a muzzle device was added.

Input Parameters

Unique Name:

308 WIN BRAKE

Cartridge:

308 Win

Firearm Class:

rifle: standard

Action Type:

bolt action

Gun Weight:

8.0

lbs

Bullet Weight:

125

grains

Muzzle Velocity:

2973

fps

Powder Charge:

46.5

grains

Muzzle Device:

brake: moderate

Stock Fit

LOP:

just right

Comb Height:

perfect cheek weld

Recoil Pad:

soft recoil pad

Calculate

Save Data Set

Results

	Data Set 1	Data Set 2	Data Set 3
Unique Name:	308 WIN	308 WIN BRAKE	--
Cartridge Selected:	308 Win	308 Win	--
Firearm Class:	rifle: standard	rifle: standard	--
Action Type:	bolt action	bolt action	--
Gun Weight:	8.0 lbs	8.0 lbs	--
Bullet Weight:	125.0 gr	125.0 gr	--
Muzzle Velocity:	2973 fps	2973 fps	--
Powder Charge:	46.5 gr	46.5 gr	--
Muzzle Device:	none	brake: moderate	--
LOP:	just right	just right	--
Comb Height:	perfect cheek weld	perfect cheek weld	--
Recoil Pad:	soft recoil pad	soft recoil pad	--

Calculated Results

Free Recoil Energy	14.92 ft-lbs	10.45 ft-lbs	-- ft-lbs
Recoil Velocity	10.96 fps	7.67 fps	-- fps
Avg Impulse Force	1434 lbf	1004 lbf	-- lbf
Perceived Recoil Score	5.7 /10	4.4 /10	-- /10



Step 8. Press the green Save Data Set button if you wish to refer back to a specific firearm – ammunition combination without having to reenter the data.

Input Parameters

Unique Name:

308 WIN BRAKE SA

Cartridge:

308 Win

Firearm Class:

rifle: standard

Action Type:

semi auto

Gun Weight:

8.0

lbs

Bullet Weight:

125

grains

Muzzle Velocity:

2973

fps

Powder Charge:

46.5

grains

Muzzle Device:

brake: moderate

Stock Fit

LOP:

just right

Comb Height:

perfect cheek weld

Recoil Pad:

soft recoil pad

Calculate

Save Data Set

Results

	Data Set 1	Data Set 2	Data Set 3
Unique Name:	308 WIN	308 WIN BRAKE	308 WIN BRAKE SA
Cartridge Selected:	308 Win	308 Win	308 Win
Firearm Class:	rifle: standard	rifle: standard	rifle: standard
Action Type:	bolt action	bolt action	semi auto
Gun Weight:	8.0 lbs	8.0 lbs	8.0 lbs
Bullet Weight:	125.0 gr	125.0 gr	125.0 gr
Muzzle Velocity:	2973 fps	2973 fps	2973 fps
Powder Charge:	46.5 gr	46.5 gr	46.5 gr
Muzzle Device:	none	brake: moderate	brake: moderate
	just right	just right	just right
	perfect cheek weld	perfect cheek weld	perfect cheek weld
	soft recoil pad	soft recoil pad	soft recoil pad

Calculated Results

Free Recoil Energy	14.92 ft-lbs	10.45 ft-lbs	10.45 ft-lbs
Recoil Velocity	10.96 fps	7.67 fps	7.67 fps
Avg Impulse Force	1434 lbf	1004 lbf	829 lbf
Perceived Recoil Score	5.7 /10	4.4 /10	3.9 /10



6. Credits & License

The Bcalc Firearm Management App was created by Brian Calc.

GNU General Public License (GPL) Version 3.